Lossless Compression Project

REPORT WRITING WORKSHEET

NOT A COMPLETED SUBMISSION REPORT. The course PDF requires the report to be written by the student. Write your own report and include a separate section titled Acknowledgments. Do not submit this worksheet unchanged.

Student(s): _____ Course: _____

1. Objective and constraints

Explain exact reconstruction, permitted imports, the CPU limit, and the compressed-data-plus-decoder-bytecode scoring metric in your own words.

2. Modeling approach

Explain the reference scheme's limits, context initialization, dictionary ordering, backoff and symbol exclusion. Describe why this models English.

3. Algorithm and file format

Explain the shared escape-estimation buckets, their initial counts and rescaling. Describe update exclusion and model count rescaling. Explain the integer range coder, carry handling and termination. Include enough detail for another student to implement a similar scheme.

Describe the header fields, alphabet, payload, output length and checksum. Explain how the decoder rebuilds all models without the original file.

4. Experiments and discussion

Use docs/benchmark_results.json and docs/validation.md as factual inputs. Record any measurements you personally repeat. Distinguish local tests from tests performed on the required ECE cluster.

Measurement	Your verified result
Original bytes and MD5	
Compressed data bytes	
Decoder bytecode bytes	
Scored total bytes	
Scored bits per character	
Decoder user CPU seconds	
Python version and machine	

Discuss context-order comparisons, memory, tradeoffs, limitations and what you learned. Do not claim global optimality or cluster validation without evidence.

Acknowledgments

REQUIRED SECTION TITLE. Attribute the main algorithmic ideas, all AI assistance, and external material actually used. State what the AI designed, implemented and tested, and what you personally did. Use docs/acknowledgments_checklist.md to check the facts.

References

List the supplied material and other sources you actually read. Verify the bibliographic details of suggested literature before citing it.

Final submission check

Submit exactly three separate files: compressed, decompress.py, and your own PDF report. Confirm exact reconstruction, CPU time below 1,200 seconds, bytecode-inclusive size, and a separate Acknowledgments section. The report must be written by you.